

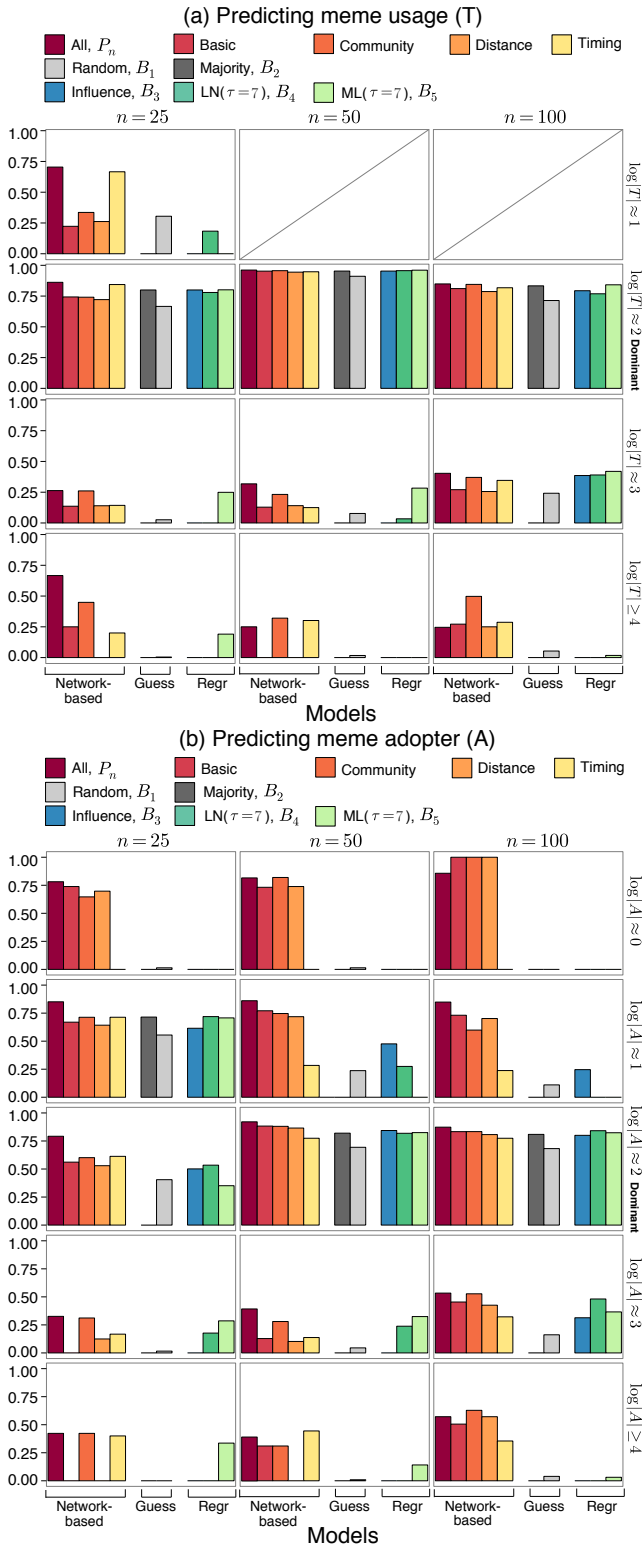


Figure 5: F_1 scores of the prediction models: (a) $\lceil \log |T| + 0.5 \rceil = 1, 2, 3, 4$ and (b) $\lceil \log |A| + 0.5 \rceil = 0, 1, 2, 3, 4$. The observation window is set to $n = 25, 50, 100$ tweets, respectively. Here we only demonstrate the results using the Infomap community detection method; link clustering yields similar results.

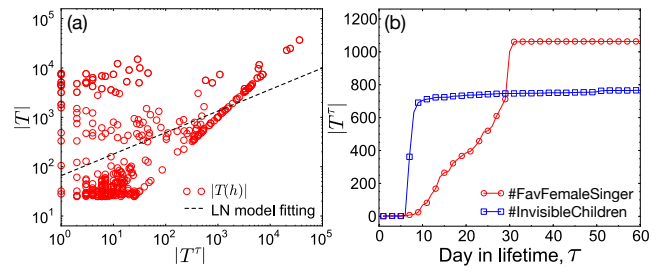


Figure 6: (a) Scatter plot of early popularity T^τ versus T for each meme; the black dashed line is the regression line by the LN model. (b) Cumulative popularity for two hashtags, #FavFemaleSinger and #InvisibleChildren.

ogy, distance, and community structure require knowledge about the network and the positions of early adopters, while the LN model, ML model, and timing features need the timestamps of early messages containing the meme. Depending on what type of information is available, one might choose different approaches.

The ability to predict whether a meme can go viral by just observing a few early messages provides us with many potential applications in social media analytics, marketing, and advertisement. This study offers not only novel, powerful features but also the first comprehensive analysis comparing multiple approaches for early prediction of viral memes.

Acknowledgement The work was supported in part by the James S. McDonnell Foundation and by NSF Grants No. 1101743 and 0910812.

Conflict of Interest Y.-Y.A. declares competing financial interests: he co-founded a social media analytics company “Jany Analytics” in 2011 and is currently one of the major shareholders. The other authors declare no competing financial interests.

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